# Poran Chowdury

**Backend Software Engineer · Java, Spring Boot, Distributed Systems, Fintech**

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## SUMMARY

Backend Software Engineer with 5+ years building distributed systems in Java and Spring Boot. Primary backend engineer of X4T, a multi-tenant crypto exchange handling real customer funds, across its identity, exchange, custody and market-data services. Strongest in retry-safe money movement, security, frontend-ready API contracts and production root-cause fixes.

## TECHNICAL SKILLS

**Backend:** Java (8–17), Spring Boot 3, Spring Data, WebFlux, Spring Batch, Hibernate/JPA, REST APIs, Node.js
**Data and Messaging:** MongoDB, PostgreSQL, SQL, Redis, RabbitMQ, Azure Service Bus, Webhooks, Server-Sent Events
**Architecture:** System Design, Microservices, Event-Driven, Multi-Tenancy, Design Patterns, Idempotency, Concurrency
**Security:** Spring Security, OAuth 2.0, JWT, Auth0, RBAC, MFA, Penetration-Test Remediation
**Cloud, DevOps and Testing:** Azure (AKS), Docker, Kubernetes, CI/CD (GitHub Actions), Linux, JUnit 5, Testcontainers, Claude Code
**Fintech and Blockchain:** Fireblocks (MPC custody), Hyperliquid, LI.FI, Circle CCTP, Web3j, EVM, Solana, Bitcoin, Tron

## PROFESSIONAL EXPERIENCE

### Software Engineer, SELISE Digital Platforms

Dhaka, Bangladesh | Jan 2022 – Present

X4T: multi-tenant crypto exchange (custody, payments, cross-chain swaps, perpetual futures); Java and Node.js on Azure AKS.

**Collaboration and AI-Assisted Delivery** 2024 – Present

- Designed the API contracts and integration guides the React team built against for perpetual futures and the live transaction-status tab (SSE), plus a dev event simulator so UI work could start early.
- Moved Hyperliquid display reads to the browser (per-user rate limits, lower latency) while the backend kept everything signed or billed, and caught a frontend polling plan that would have exhausted the platform's API budget.
- Sped up delivery with agentic development (Claude Code, MCP, parallel agents in git worktrees), with every change verified by tests and code review before merge. Merged 98 PRs across 3 repos in 7 weeks.

**Trading and Custody** 2024 – Present

- Took perpetual-futures trading on Hyperliquid to production in ~3 weeks across Java and Node.js: a hexagonal core with ~1,700 tests, fail-fast pre-trade risk checks, order signing through Fireblocks, and reconciliation against the exchange.
- Enabled Bitcoin, Solana and Tron trading on LI.FI via pluggable chain adapters, filtering out routes that would fail on submission.
- Let small-balance users sell their full balance by replacing flat gas buffers (e.g. ~$20 for BTC) with live per-chain fees.
- Designed Visa card issuing on Reap: a double-entry ledger, sagas across Fireblocks and Reap, an outbox and reconciliation.
- Made fund movements retry-safe across services, Fireblocks and webhooks with end-to-end idempotency keys, owner-checked Redis locks and terminal-state guards. Fixed a production race condition that created duplicate custody vaults.

**Identity, Security and Compliance** 2022 – Present

- Built the multi-tenant identity service on Auth0 from its first release (tenant resolution, MFA, password reset, STS token exchange, RabbitMQ events); later added Google and Apple SSO, device sessions and Auth0 tenant isolation.
- Owned the Sumsub-to-Didit KYC/KYB migration end to end, moving 2,366 approved users with documents and liveness video, with no re-verification and no user notifications.
- Led penetration-test remediation: OTP step-up for withdrawals, IDOR and NoSQL-injection fixes, server-side fees, a CORS allow-list, and signup-abuse controls (Cloudflare Turnstile, disposable-email blocklist, rate limiting).

**Platform and Reliability** 2022 – Present

- Built the shared Java platform library every service depends on: per-request database-per-tenant MongoDB routing, per-tenant JWT validation, and RabbitMQ topology with dead-letter queues.
- Added page-size caps and per-key Redis rate limiting (429) after bulk partner-API queries caused an out-of-memory outage.
- Improved market-data performance: a 10-minute refresh cut to 2 with Spring WebFlux, a contract-address index and batching.
- Migrated the identity service to Spring Boot 3.5 with 570+ tests, an 80% coverage gate and Testcontainers end-to-end tests.
- Urstamm (Swiss timber traceability, 2023-24): Spring Boot 3 migration, 10,000+ records on-chain via Spring Batch and Web3j.

### Java Developer, SELISE Digital Platforms

Dhaka, Bangladesh | May 2021 – Jan 2022

- Developed Spring Boot REST APIs with Spring Data and third-party API integrations.

## PROJECTS

**Blocks LMT Java Client** | Spring Boot, OpenTelemetry, Azure Service Bus Maven Central: com.seliseblocks.core:blocks-lmt-java-client

- Observability starter that auto-configures OpenTelemetry and ships logs and spans to Azure Service Bus; used by 4 services.

## EDUCATION

**B.Sc. in Computer Science and Engineering, Southeast University** Dhaka, Bangladesh | 2017 – 2022